



Avoid stressful driving

Science says to avoid driving while stressed. Driving to work is the most stressed out task. <http://dgit.in/GettoWork>



Know Steve Jobs

We all knew Steve Jobs on the stage, but did we really know him? <http://dgit.in/KnowJobs>



Intel in the house

We got a chance to catch up with Anuj Dua, at the Intel 6th Generation Experience Summit held at the sprawling Sydney Opera House. Skylake was center stage and made for some interesting conversation.

◀ Anuj Dua, Director for Marketing, Intel Asia Pacific Japan

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Talking about Skylake, we tested the 6700K recently and noticed that over Devil's Canyon (4790K) the performance upgrade was about five percent. Do you think we are hitting the proverbial brick wall when it comes to performance?

Anuj: No, not really. Actually for the 'K' series, in the desktop we basically skipped a generation of upgrade. We had a big jump with Haswell. Skylake, from a micro-architecture standpoint is re-architected. It's a "tock" as you know, and it was built to be scalable and give you a big performance boost with a focus on the media engine – 4k playback etc. If you look at the notebook space, there is a double digit jump in performance, between generation to generation. When you think of the last product of an old architecture and the first product of a new architecture, typically the delta isn't that great. If you compare end of life to end of life, there is a huge gain. So think of it as you start off the chute with an improvement and over time as you start you get the process technology figured out. At the end of the day, these gaming SKUs are all about clocking, so having faster IPC – instructions per clock gain – is one piece of it, but as the GHz start going up,

as the process technology starts maturing, as you implement a lot of circuit fixes, the GHz gives you a big benefit in terms of gaming performance. So, out shoots a very mature architecture that is end of life and the new architecture having a 5-10 percent gain in the gaming series, that doesn't surprise me.

Right, like you said it's a "tock" so the process is still the 14nm Broadwell process. Ivybridge brought in the tri-gate 3D transistors to enable a shrinkage. What will be the next technological breakthrough which will change things like the three transistors did?

Anuj: One thing I can tell you is that there is a lot of physics and material science research involved at Intel in continuing the Moore's Law as projected. We've always said that we never know the roadmap beyond 10 years. We have visibility for 10 years at any given point of time and even at this point of time, we have visibility till 10 years as well. Through it all it requires reinvention of the transistor structure, the dielectric, the materials that are used and a host of different things. Who had ever thought or had a notion that a 3D transistor would come to life right? So, I can't tell you what exactly is planned. Part of it is that I don't know all the details, and the other is even

if I knew, I wouldn't be in a position to tell you. But that's essentially the IP that we have. People who are involved in the R&D at Intel, invest a lot of time and effort and money to find those breakthroughs. And so we will see the evolution that plays out but that's certainly part of the 14nm maturing process.

You said that a lot of R&D is going on and you guys are trying to approach the problem from the efficiency point of view. The rationale is that in order to make our devices last longer the processors are becoming more and more efficient. In effect you are hacking away at the silicon but does Intel by any chance do any R&D into battery tech?

Anuj: Oh sure, we have a microprocessor technology team, we also have an eco-system and an infrastructure technology team that looks at everything from cables to battery technologies to camera technology and all those things. When you think about something like Type-C connectors, and the thunderbolt connectors – those all came out of research at Intel. Because at the end of the day of you think about it, your microprocessor is becoming very fast, now the bottleneck has moved to memory subsystem. So if you don't have a well balanced memory plus processor capability, you end up crunching up all